Software Engineering & Project Management (SEPM)

Energy & Utility Management System (EUMS)

Billing, metering and customer service for electricity, water and gas

User Interface Screens

Software process model: Waterfall · Combined SEPM + DBMS course project

Group number	Group 3
Course and section	Software Engineering & Project Management (SEPM), CSEF
Date	October 2026

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1. User Interface Screens

The prototype is the working application itself. Each feature has its own page. Admin and employee pages share the staff portal (employees do not see Employees, Tariffs and Reports); customers have a separate portal.

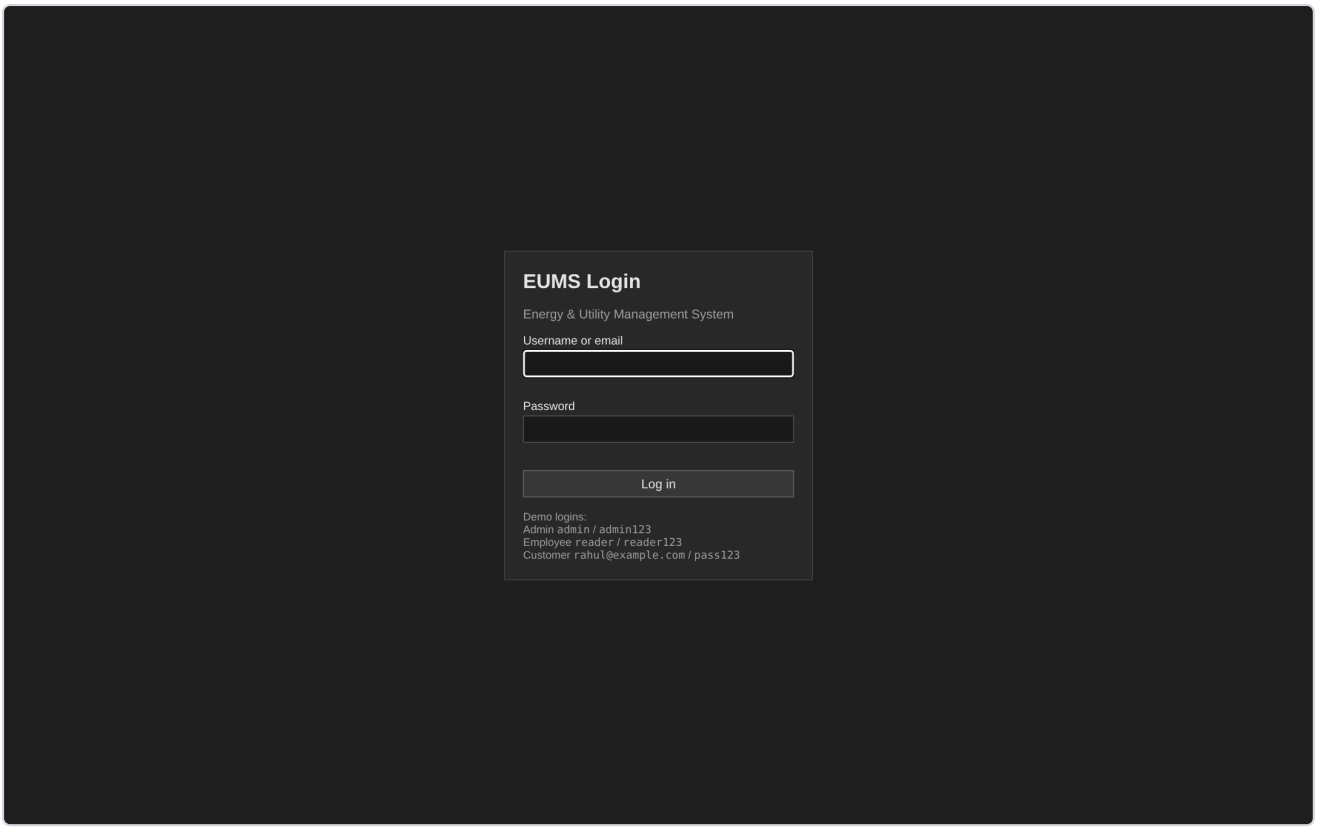


Figure 1: Login page with demo accounts

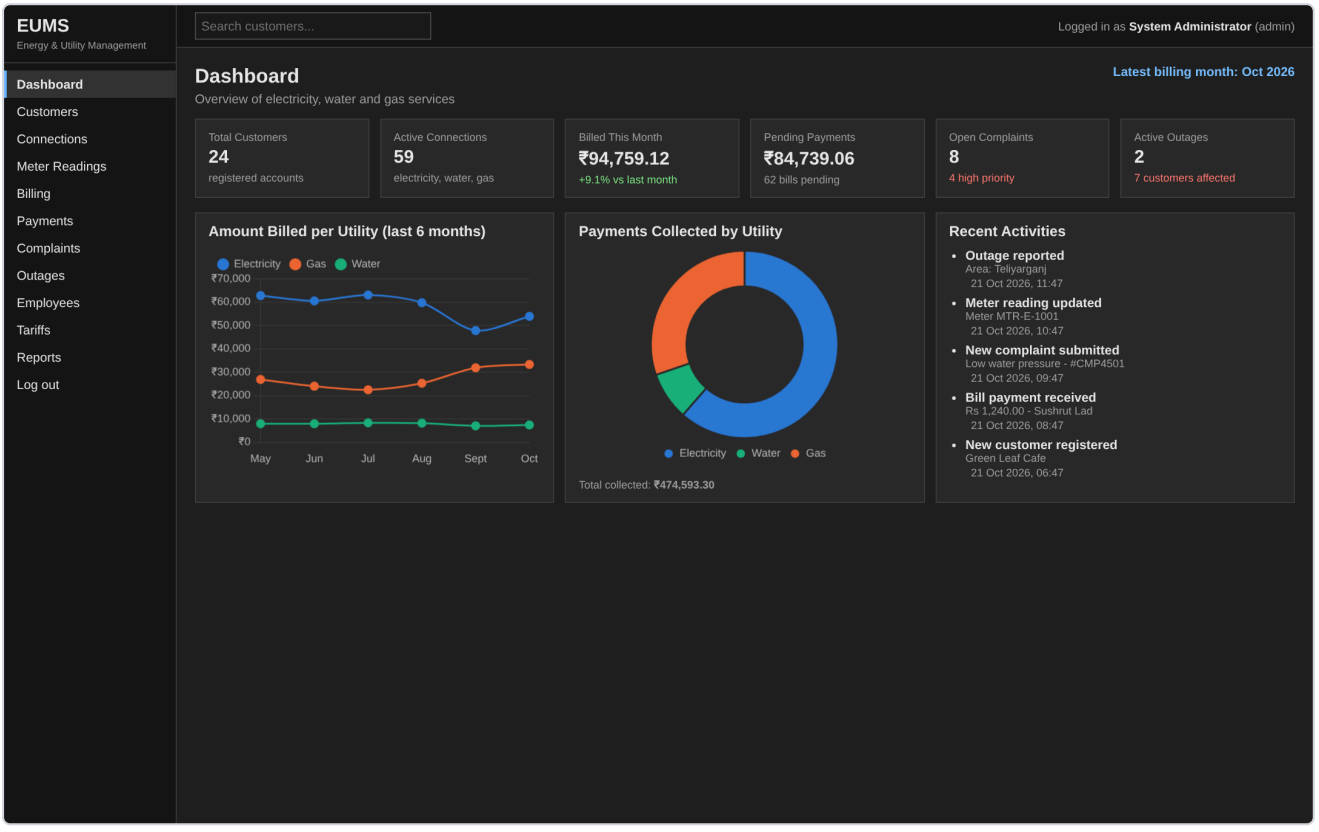


Figure 2: Admin dashboard: KPIs, billed amount per utility, collections, recent activity

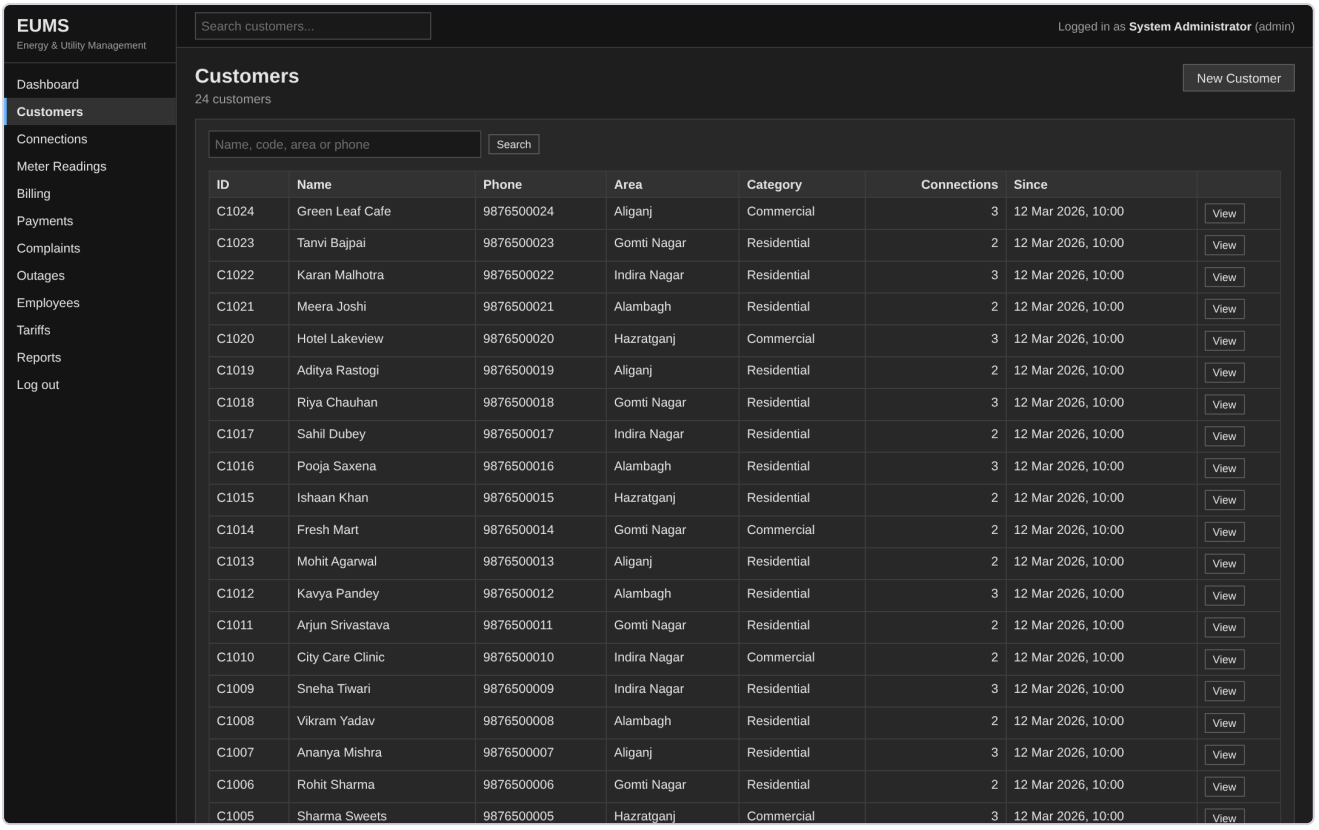


Figure 3: Customers list with search

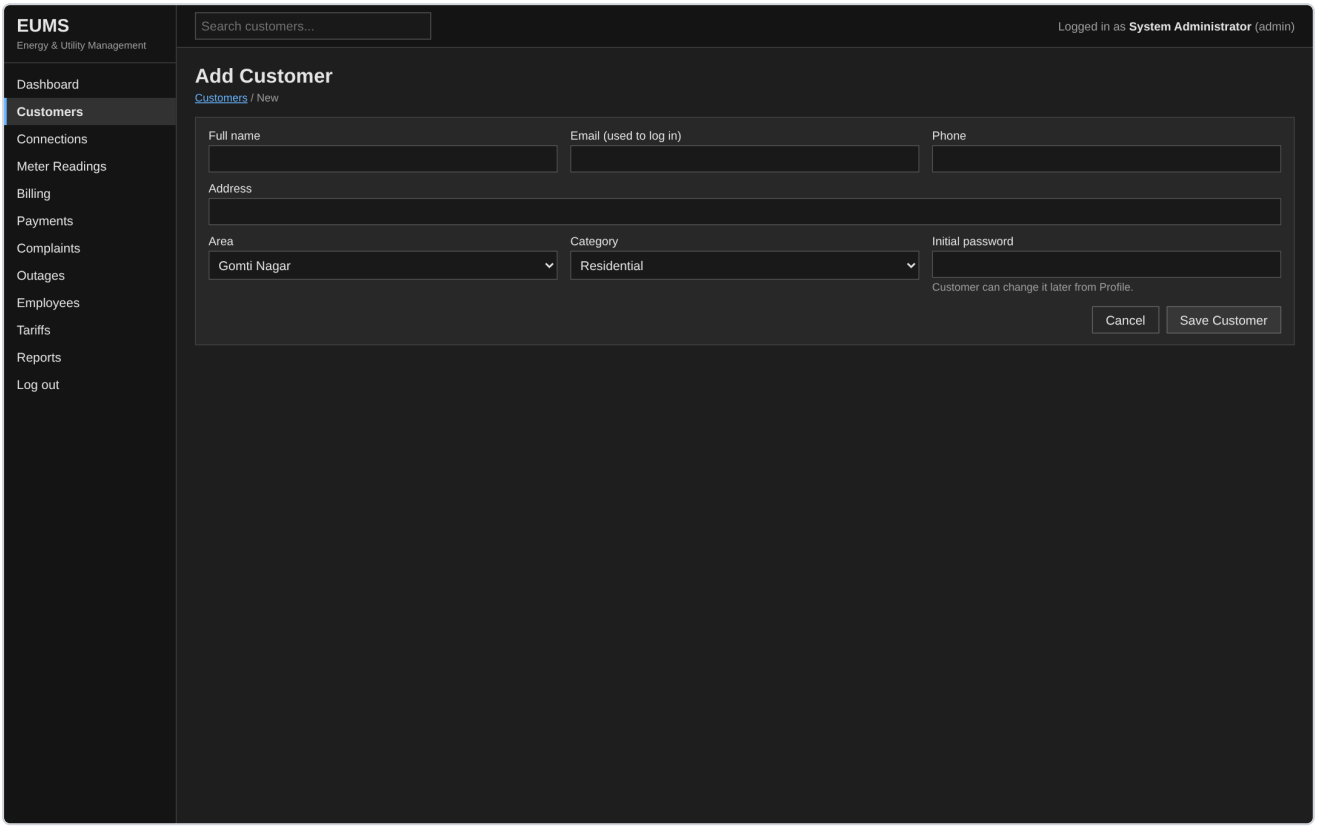


Figure 4: Add customer form

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Search customers...

Logged in as **System Administrator** (admin)

- Dashboard
- Customers
- Connections
- Meter Readings
- Billing
- Payments
- Complaints
- Outages
- Employees
- Tariffs
- Reports
- Log out

Connections

59 connections

New Connection

All | Electricity | Water | Gas

Connection	Customer	Utility	Tariff	Meter	Area	Installed	Status
GAS1059	Green Leaf Cafe C1024	Gas	Commercial PNG	MTR-G-1059	Aliganj	12 Mar 2026	Active
WATE1058	Green Leaf Cafe C1024	Water	Commercial Water	MTR-W-1058	Aliganj	12 Mar 2026	Active
ELEC1057	Green Leaf Cafe C1024	Electricity	Commercial Electricity (LMV-2)	MTR-E-1057	Aliganj	12 Mar 2026	Active
WATE1056	Tanvi Bajpai C1023	Water	Domestic Water	MTR-W-1056	Gomti Nagar	12 Mar 2026	Active
ELEC1055	Tanvi Bajpai C1023	Electricity	Domestic Electricity (LMV-1)	MTR-E-1055	Gomti Nagar	12 Mar 2026	Active
GAS1054	Karan Malhotra C1022	Gas	Domestic PNG	MTR-G-1054	Indira Nagar	12 Mar 2026	Active
WATE1053	Karan Malhotra C1022	Water	Domestic Water	MTR-W-1053	Indira Nagar	12 Mar 2026	Active
ELEC1052	Karan Malhotra C1022	Electricity	Domestic Electricity (LMV-1)	MTR-E-1052	Indira Nagar	12 Mar 2026	Active
WATE1051	Meera Joshi C1021	Water	Domestic Water	MTR-W-1051	Alambagh	12 Mar 2026	Active
ELEC1050	Meera Joshi C1021	Electricity	Domestic Electricity (LMV-1)	MTR-E-1050	Alambagh	12 Mar 2026	Active
GAS1049	Hotel Lakeview C1020	Gas	Commercial PNG	MTR-G-1049	Hazratganj	12 Mar 2026	Active
WATE1048	Hotel Lakeview C1020	Water	Commercial Water	MTR-W-1048	Hazratganj	12 Mar 2026	Active
ELEC1047	Hotel Lakeview C1020	Electricity	Commercial Electricity (LMV-2)	MTR-E-1047	Hazratganj	12 Mar 2026	Active
GAS1046	Aditya Rastogi C1019	Gas	Domestic PNG	MTR-G-1046	Aliganj	12 Mar 2026	Active
ELEC1045	Aditya Rastogi C1019	Electricity	Domestic Electricity (LMV-1)	MTR-E-1045	Aliganj	12 Mar 2026	Active
GAS1044	Riya Chauhan C1018	Gas	Domestic PNG	MTR-G-1044	Gomti Nagar	12 Mar 2026	Active
WATE1043	Riya Chauhan C1018	Water	Domestic Water	MTR-W-1043	Gomti Nagar	12 Mar 2026	Active
ELEC1042	Riya Chauhan C1018	Electricity	Domestic Electricity (LMV-1)	MTR-E-1042	Gomti Nagar	12 Mar 2026	Active
WATE1041	Sahil Dubey C1017	Water	Domestic Water	MTR-W-1041	Indira Nagar	12 Mar 2026	Active
ELEC1040	Sahil Dubey C1017	Electricity	Domestic Electricity (LMV-1)	MTR-E-1040	Indira Nagar	12 Mar 2026	Active
GAS1039	Pooja Saxena C1016	Gas	Domestic PNG	MTR-G-1039	Alambagh	12 Mar 2026	Active
WATE1038	Pooja Saxena C1016	Water	Domestic Water	MTR-W-1038	Alambagh	12 Mar 2026	Active
ELEC1037	Pooja Saxena C1016	Electricity	Domestic Electricity (LMV-1)	MTR-E-1037	Alambagh	12 Mar 2026	Active
GAS1036	Ishaan Khan C1015	Gas	Domestic PNG	MTR-G-1036	Hazratganj	12 Mar 2026	Active

Figure 5: Connections, filterable by utility

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Search customers...

Logged in as **System Administrator** (admin)

- Dashboard
- Customers
- Connections
- Meter Readings
- Billing
- Payments
- Complaints
- Outages
- Employees
- Tariffs
- Reports
- Log out

Meter Readings

Latest 100 readings

Add Reading

Date	Meter	Customer	Utility	Previous	Current	Consumption	Recorded by
02 Oct 2026	MTR-G-1059	Green Leaf Cafe	Gas	558.3	629.8	71.5 SCM	Suresh Yadav
02 Oct 2026	MTR-W-1058	Green Leaf Cafe	Water	868.5	919.3	50.8 kL	Suresh Yadav
02 Oct 2026	MTR-E-1057	Green Leaf Cafe	Electricity	7004.8	7779.2	774.4 kWh	Suresh Yadav
02 Oct 2026	MTR-W-1056	Tanvi Bajpai	Water	647.6	661.3	13.7 kL	Suresh Yadav
02 Oct 2026	MTR-E-1055	Tanvi Bajpai	Electricity	11324.5	11544.5	220.0 kWh	Suresh Yadav
02 Oct 2026	MTR-G-1054	Karan Malhotra	Gas	958.6	990.3	31.7 SCM	Suresh Yadav
02 Oct 2026	MTR-W-1053	Karan Malhotra	Water	346.6	369.3	22.7 kL	Suresh Yadav
02 Oct 2026	MTR-E-1052	Karan Malhotra	Electricity	5387.4	5591.8	204.4 kWh	Suresh Yadav
02 Oct 2026	MTR-W-1051	Meera Joshi	Water	441.8	451.6	9.8 kL	Suresh Yadav
02 Oct 2026	MTR-E-1050	Meera Joshi	Electricity	7544.8	7833.0	288.2 kWh	Suresh Yadav
02 Oct 2026	MTR-G-1049	Hotel Lakeview	Gas	935.7	1009.4	73.7 SCM	Suresh Yadav
02 Oct 2026	MTR-W-1048	Hotel Lakeview	Water	535.4	583.3	47.9 kL	Suresh Yadav
02 Oct 2026	MTR-E-1047	Hotel Lakeview	Electricity	6364.9	7207.3	842.4 kWh	Suresh Yadav
02 Oct 2026	MTR-G-1046	Aditya Rastogi	Gas	365.0	386.4	21.4 SCM	Suresh Yadav
02 Oct 2026	MTR-E-1045	Aditya Rastogi	Electricity	3801.7	4020.9	219.2 kWh	Suresh Yadav
02 Oct 2026	MTR-G-1044	Riya Chauhan	Gas	390.1	415.9	25.8 SCM	Suresh Yadav
02 Oct 2026	MTR-W-1043	Riya Chauhan	Water	295.4	311.5	16.1 kL	Suresh Yadav
02 Oct 2026	MTR-E-1042	Riya Chauhan	Electricity	3232.6	3536.4	303.8 kWh	Suresh Yadav
02 Oct 2026	MTR-W-1041	Sahil Dubey	Water	729.5	745.7	16.2 kL	Suresh Yadav
02 Oct 2026	MTR-E-1040	Sahil Dubey	Electricity	10322.3	10463.8	141.5 kWh	Suresh Yadav
02 Oct 2026	MTR-G-1039	Pooja Saxena	Gas	881.7	913.8	32.1 SCM	Suresh Yadav
02 Oct 2026	MTR-W-1038	Pooja Saxena	Water	648.3	658.4	10.1 kL	Suresh Yadav
02 Oct 2026	MTR-E-1037	Pooja Saxena	Electricity	14159.7	14413.0	253.3 kWh	Suresh Yadav
02 Oct 2026	MTR-G-1036	Ishaan Khan	Gas	617.0	644.2	27.2 SCM	Suresh Yadav
02 Oct 2026	MTR-E-1035	Ishaan Khan	Electricity	7809.8	8008.7	198.9 kWh	Suresh Yadav

Figure 6: Meter readings list

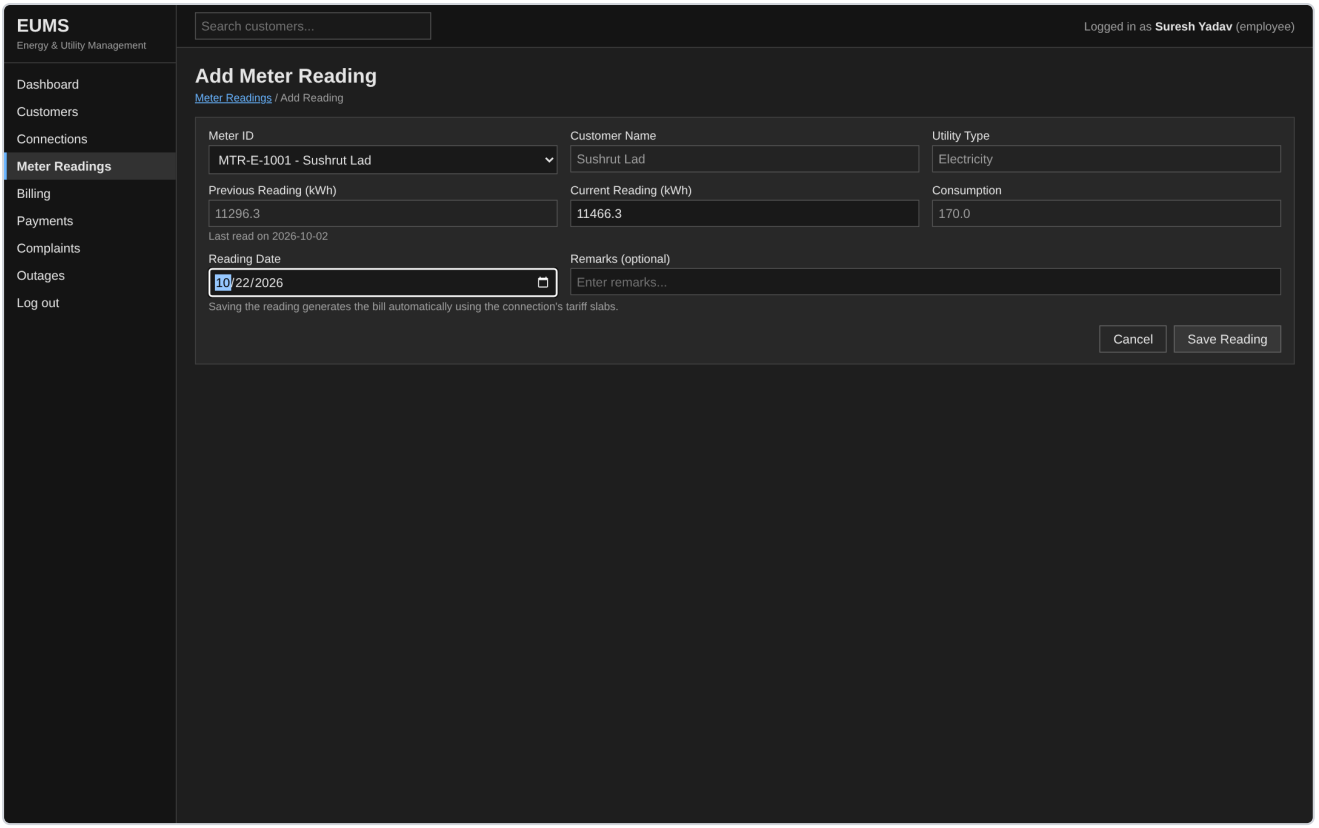


Figure 7: Add meter reading (employee)

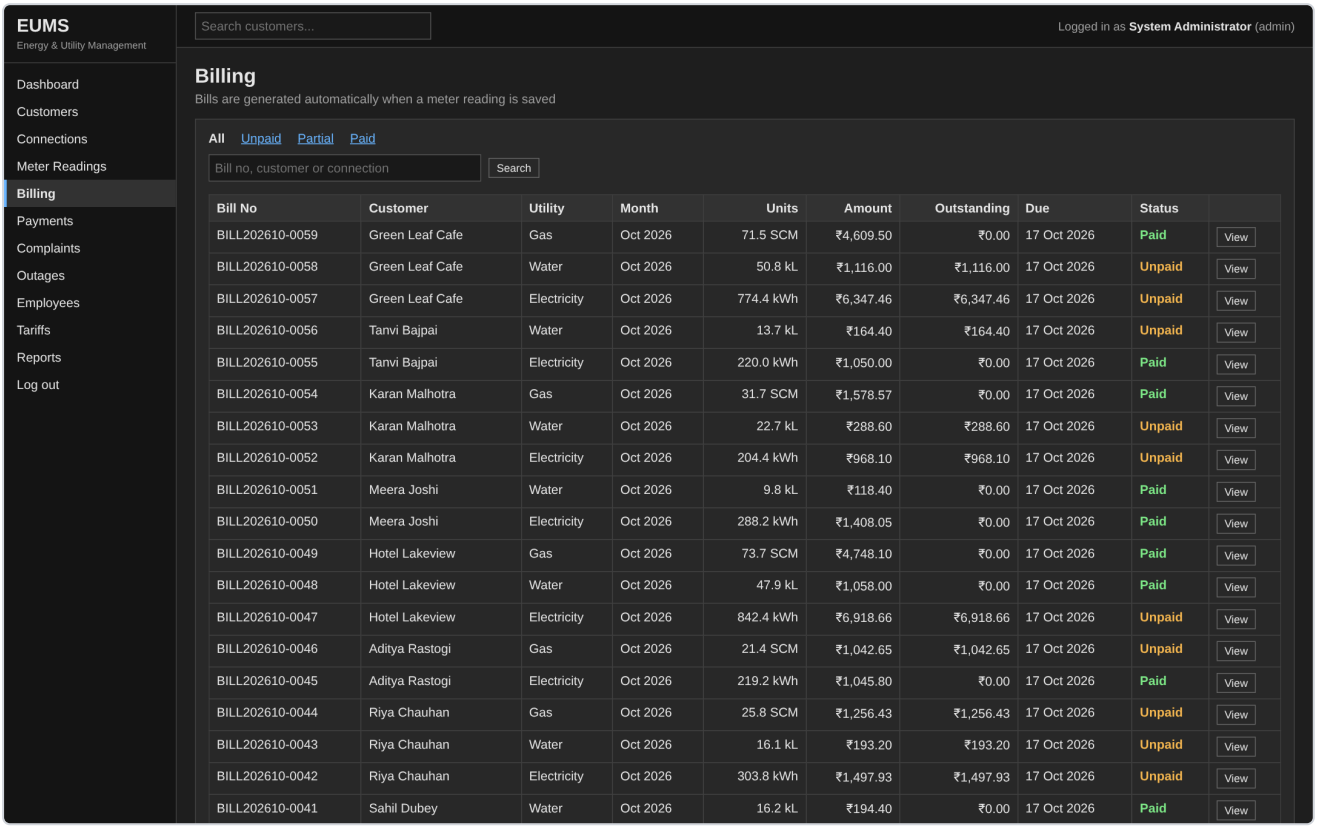


Figure 8: Billing list with status tabs

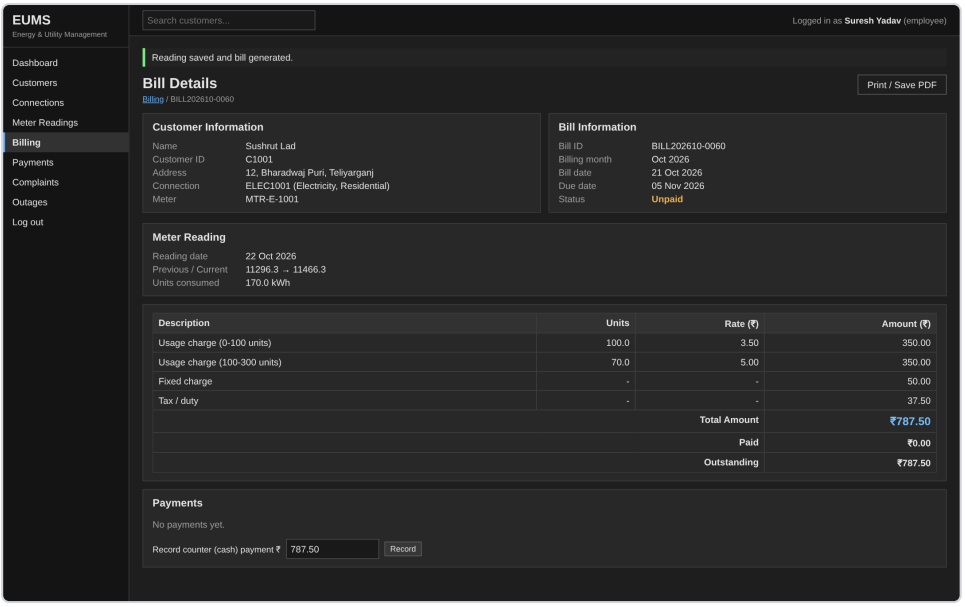


Figure 9: Bill details with slab breakdown and counter payment

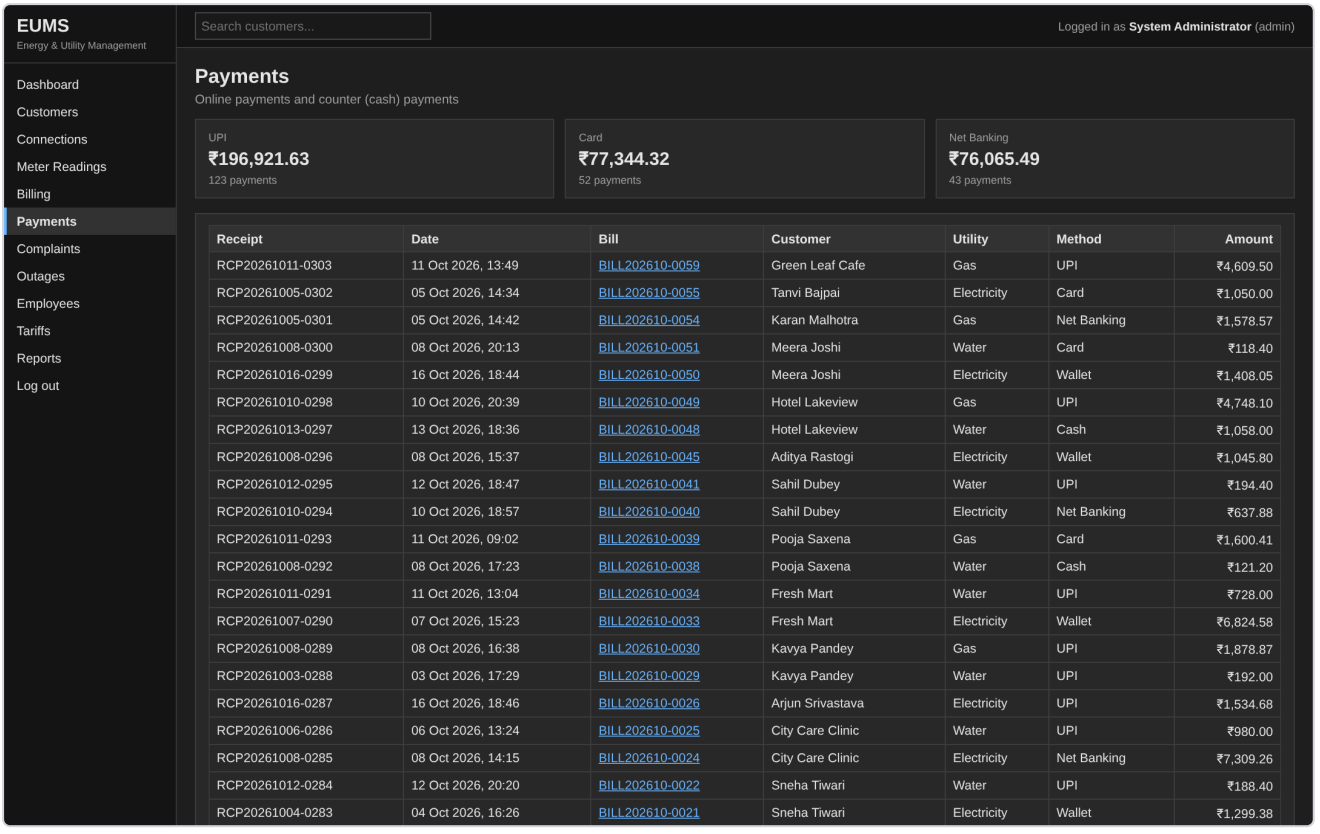


Figure 10: Payments with totals by method

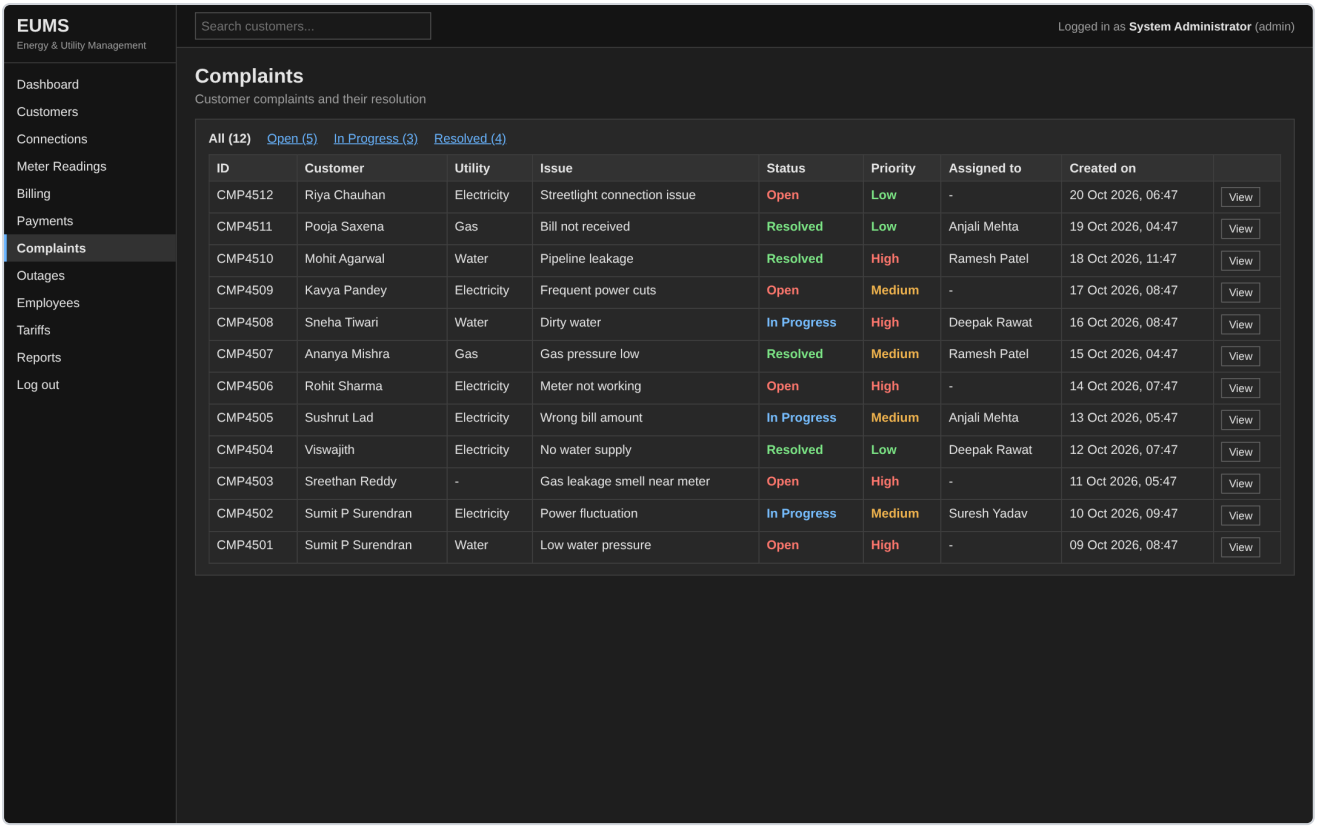


Figure 11: Complaint management

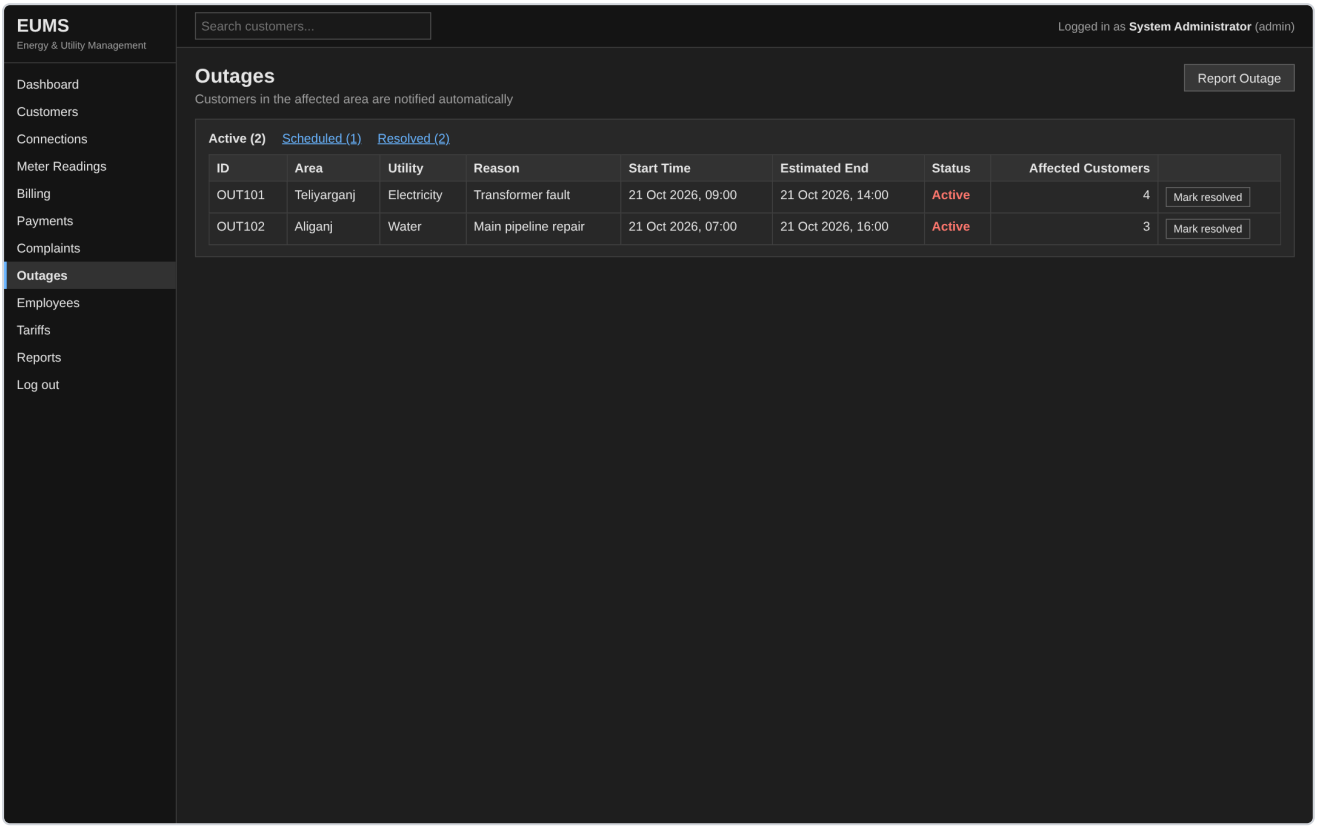


Figure 12: Outage management

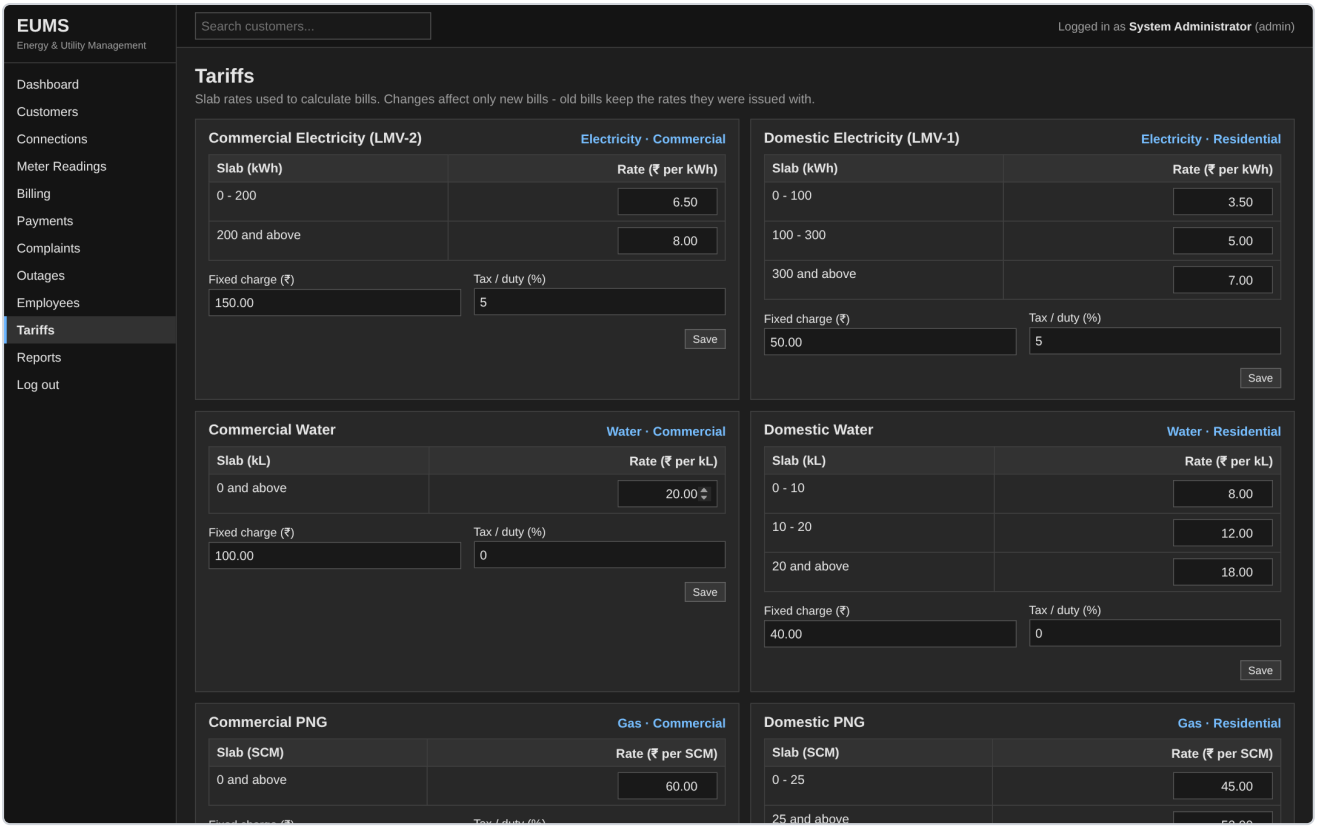


Figure 13: Tariff editor (admin)

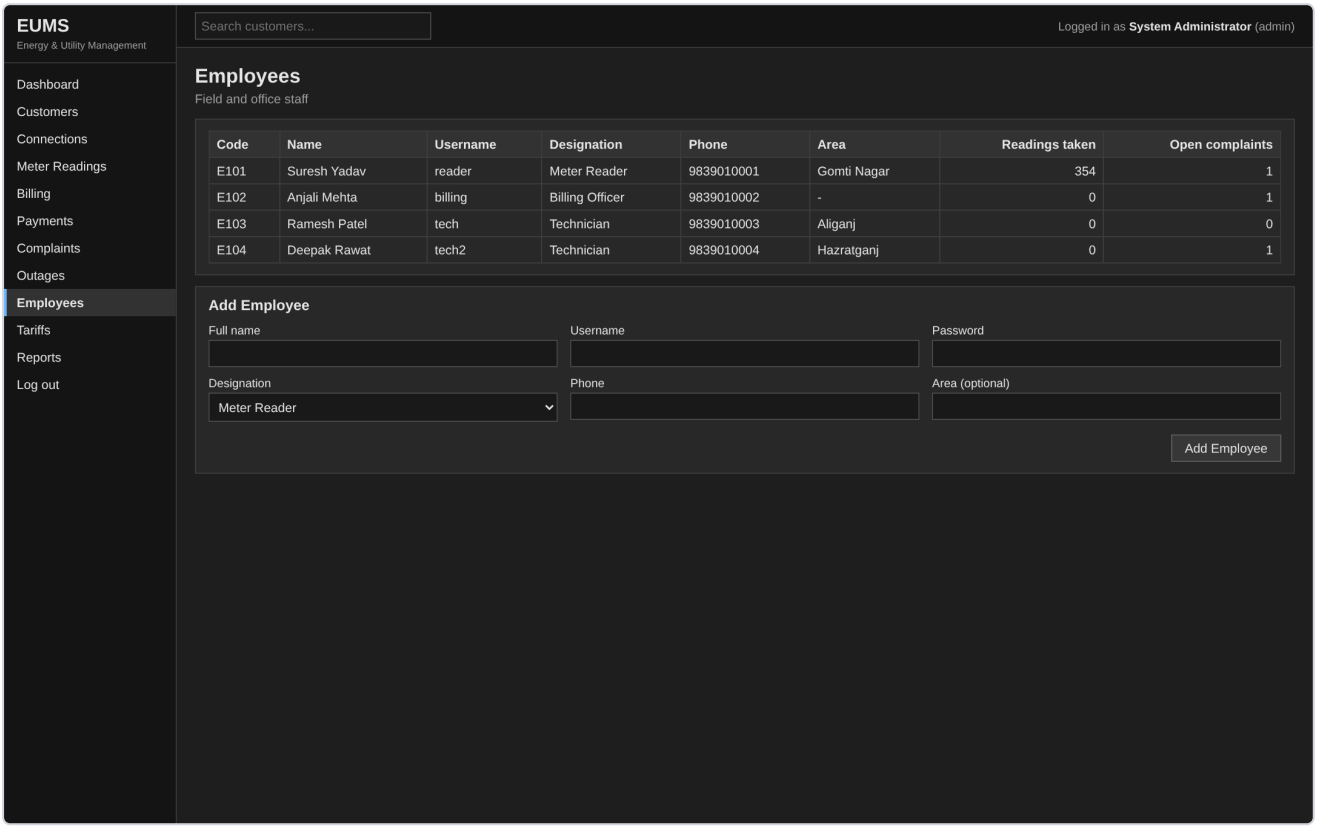


Figure 14: Employees (admin)

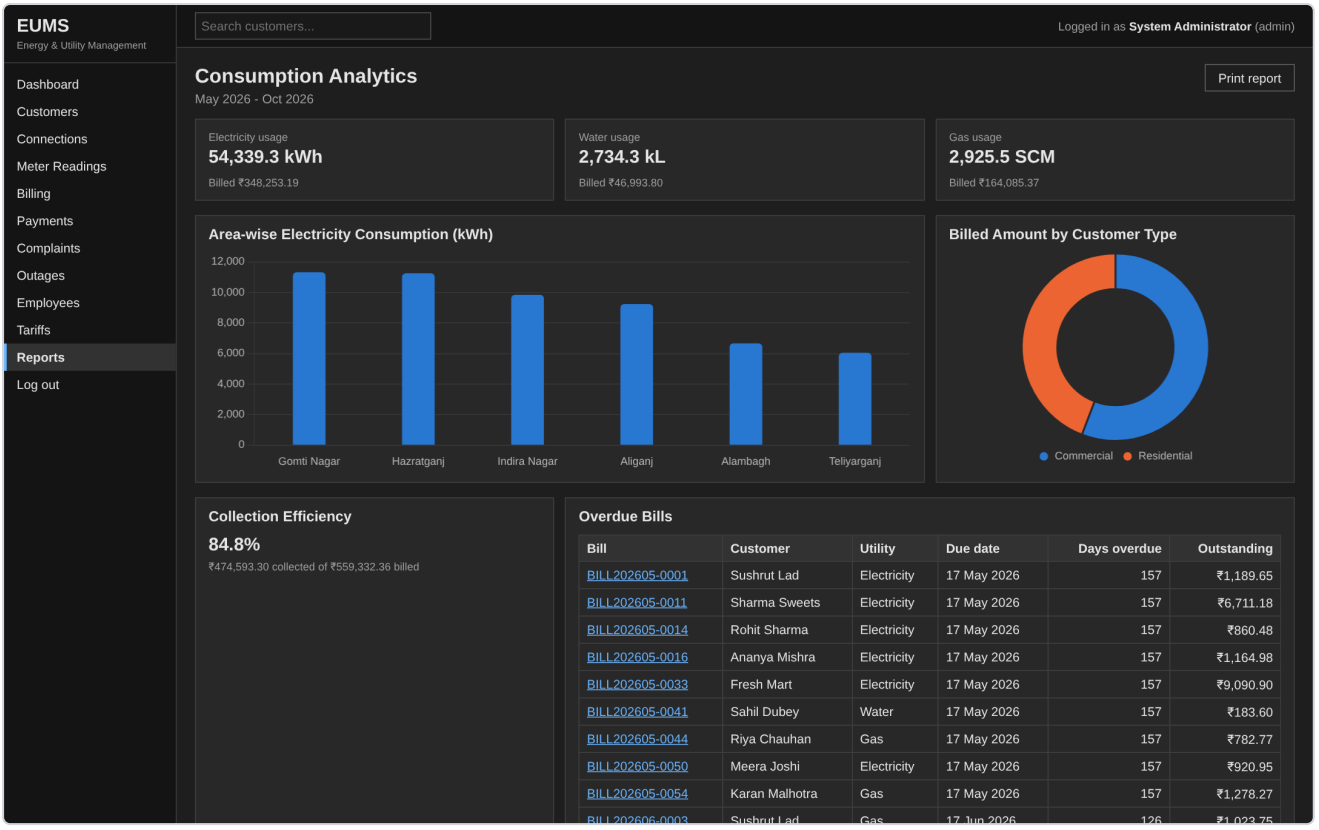


Figure 15: Consumption analytics and overdue bills (admin)

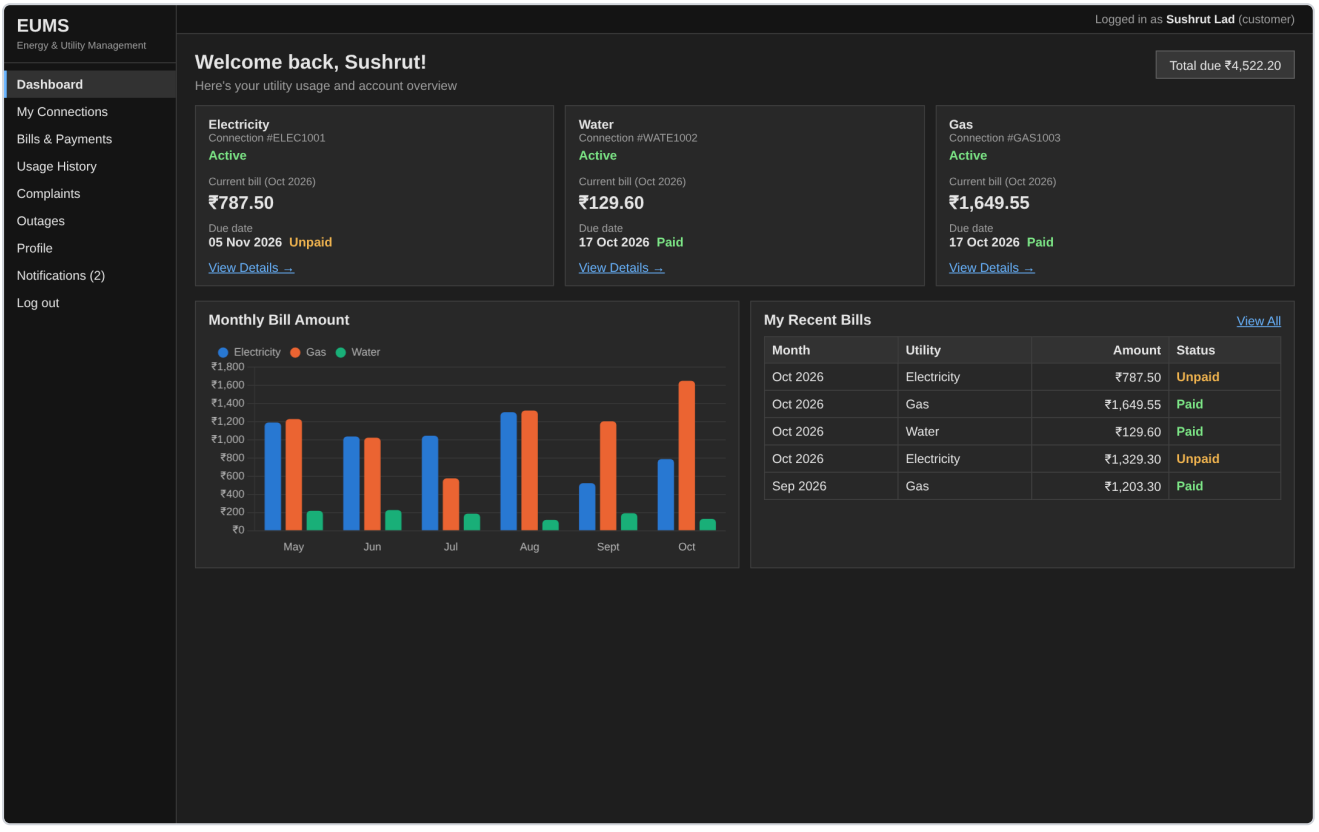


Figure 16: Customer dashboard

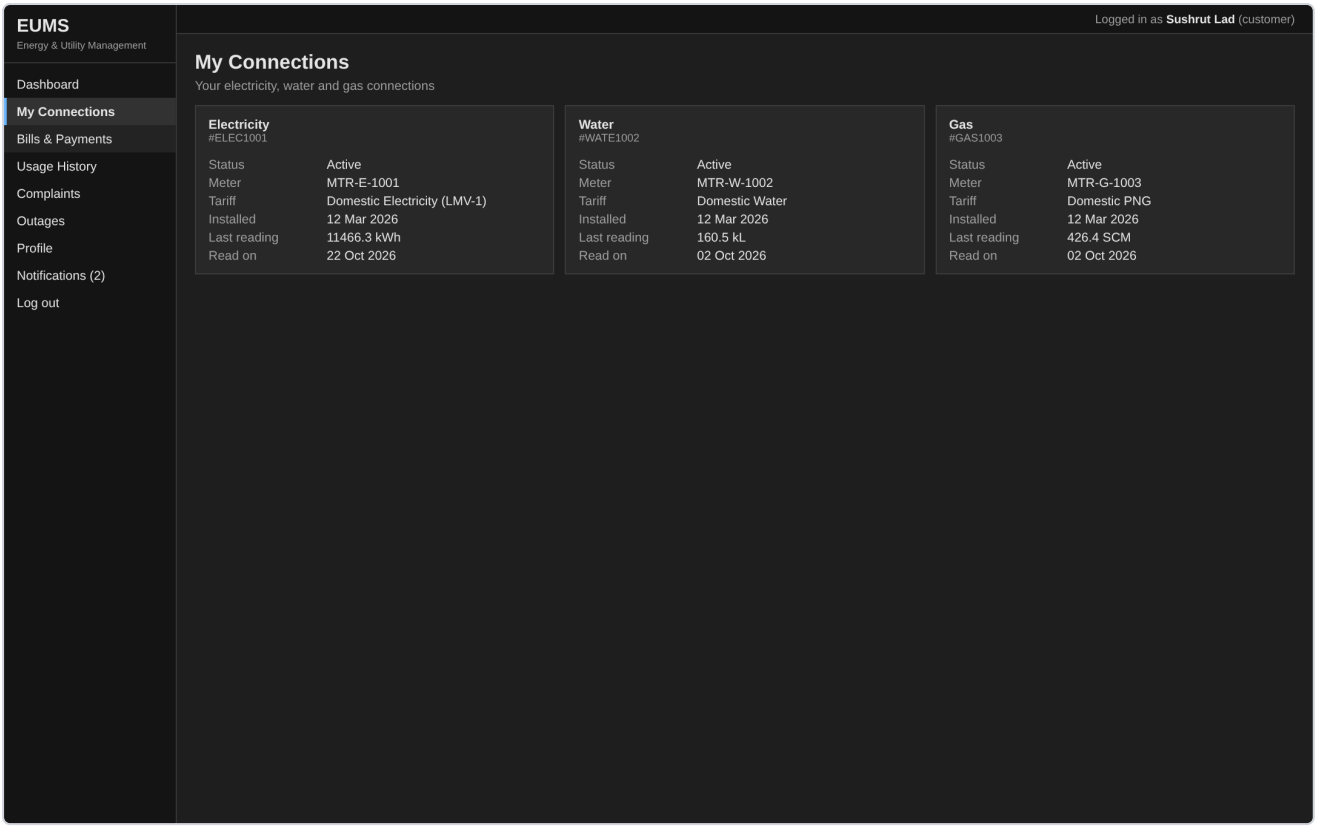


Figure 17: Customer: my connections

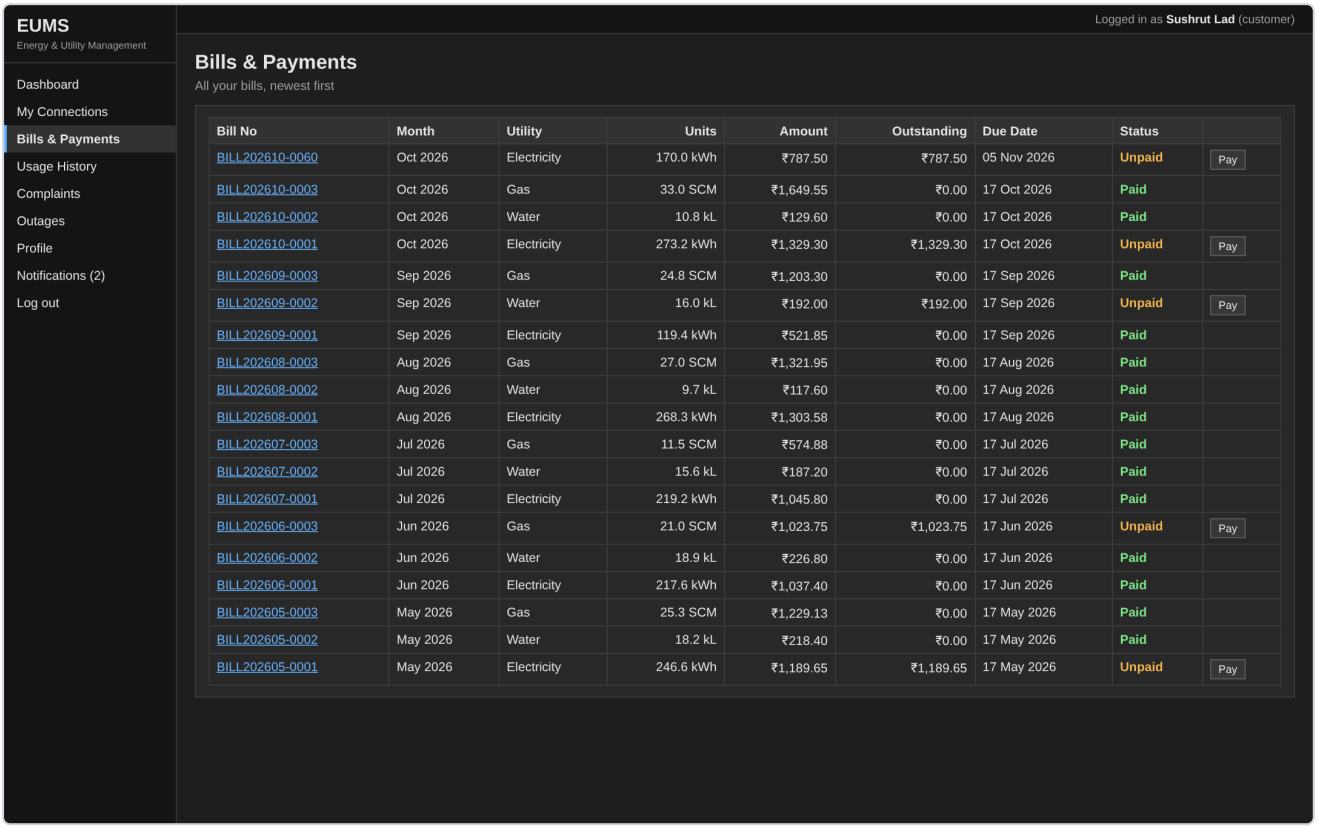


Figure 18: Customer: bills and payments

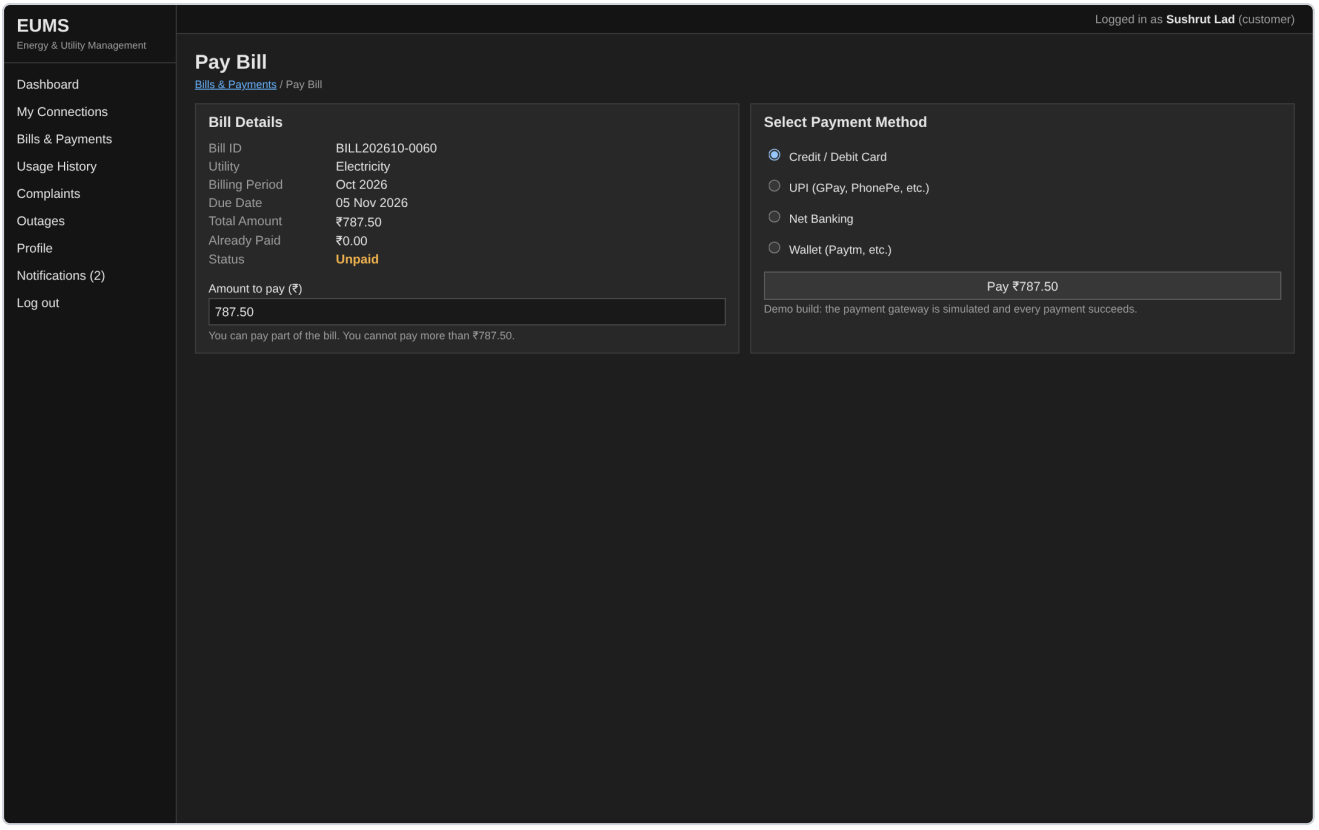


Figure 19: Customer: pay bill

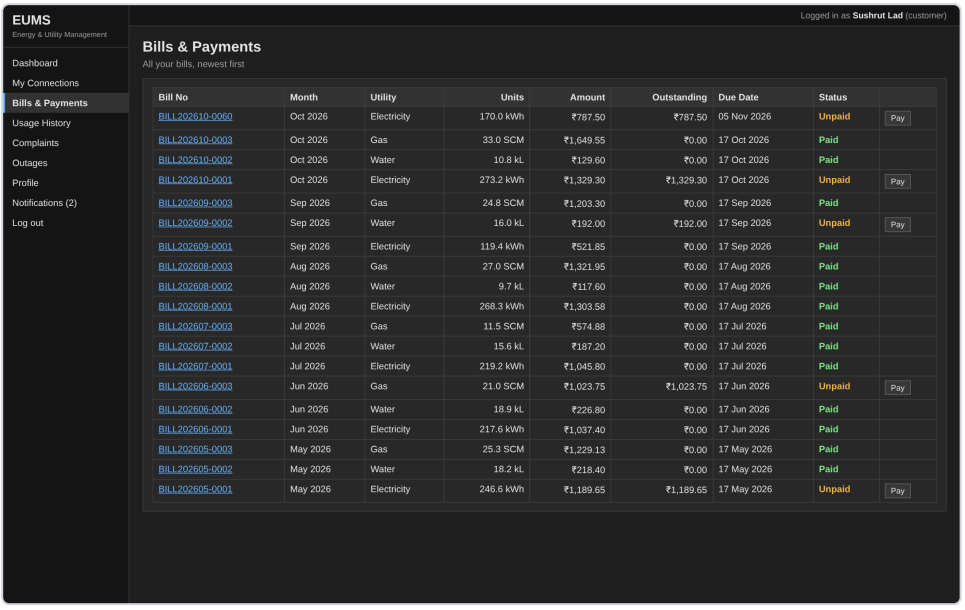


Figure 20: Customer: bill after successful payment

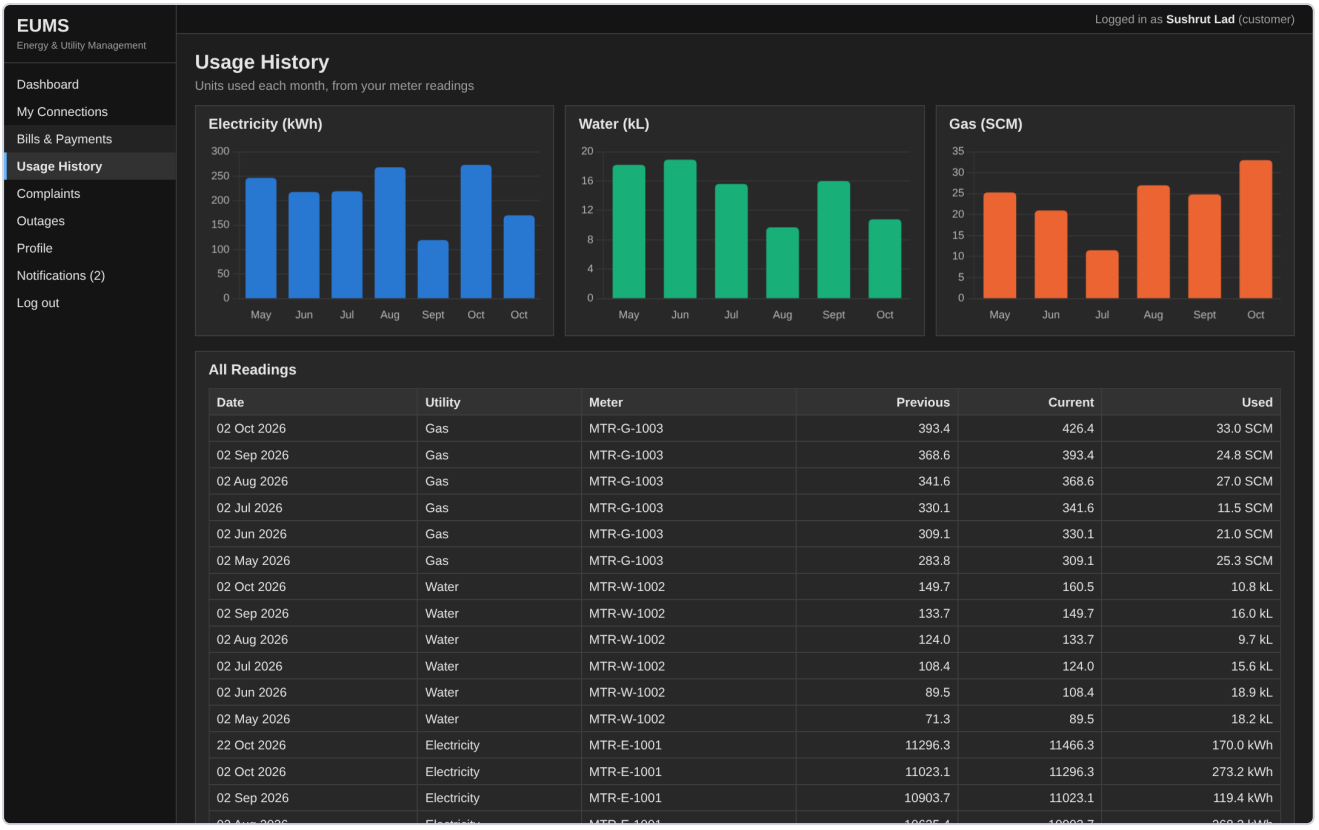


Figure 21: Customer: usage history per utility

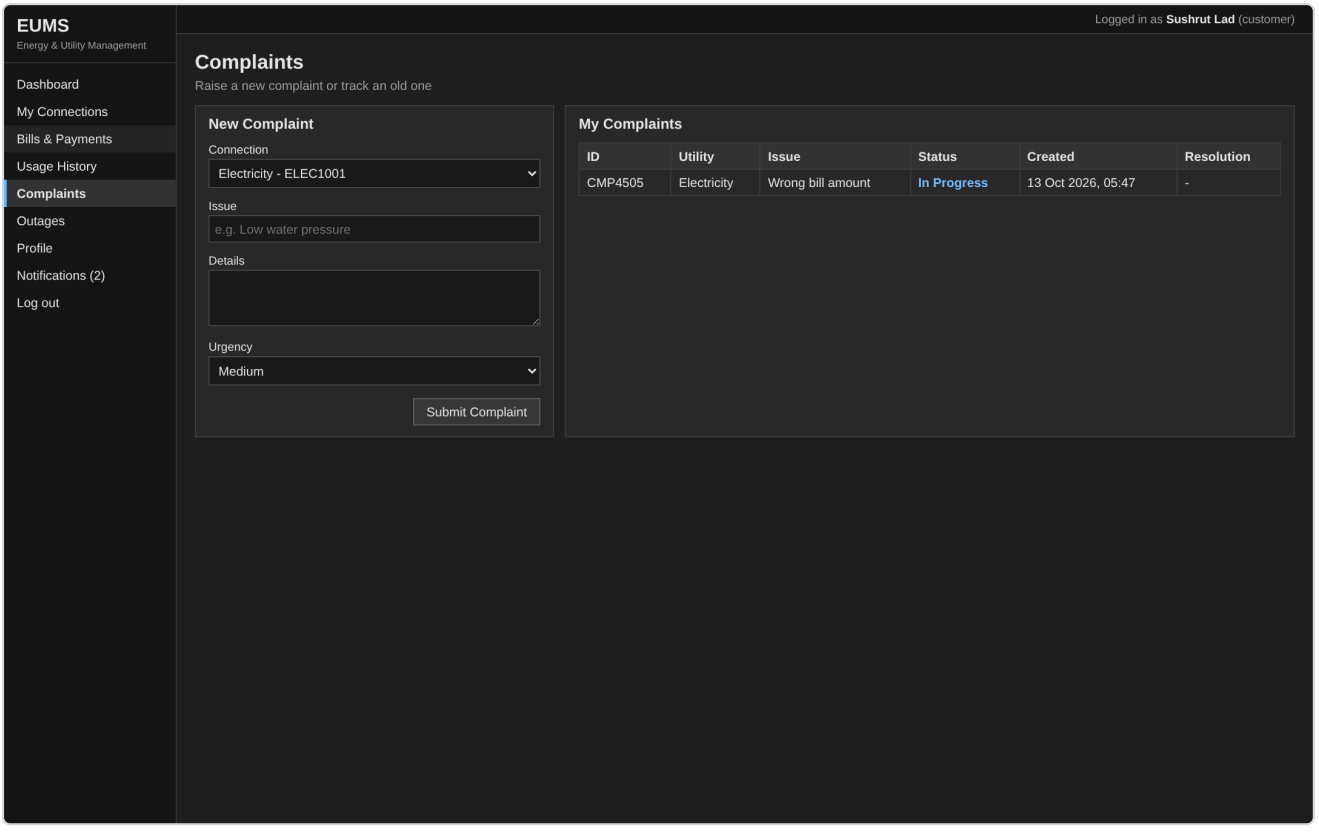


Figure 22: Customer: raise and track complaints

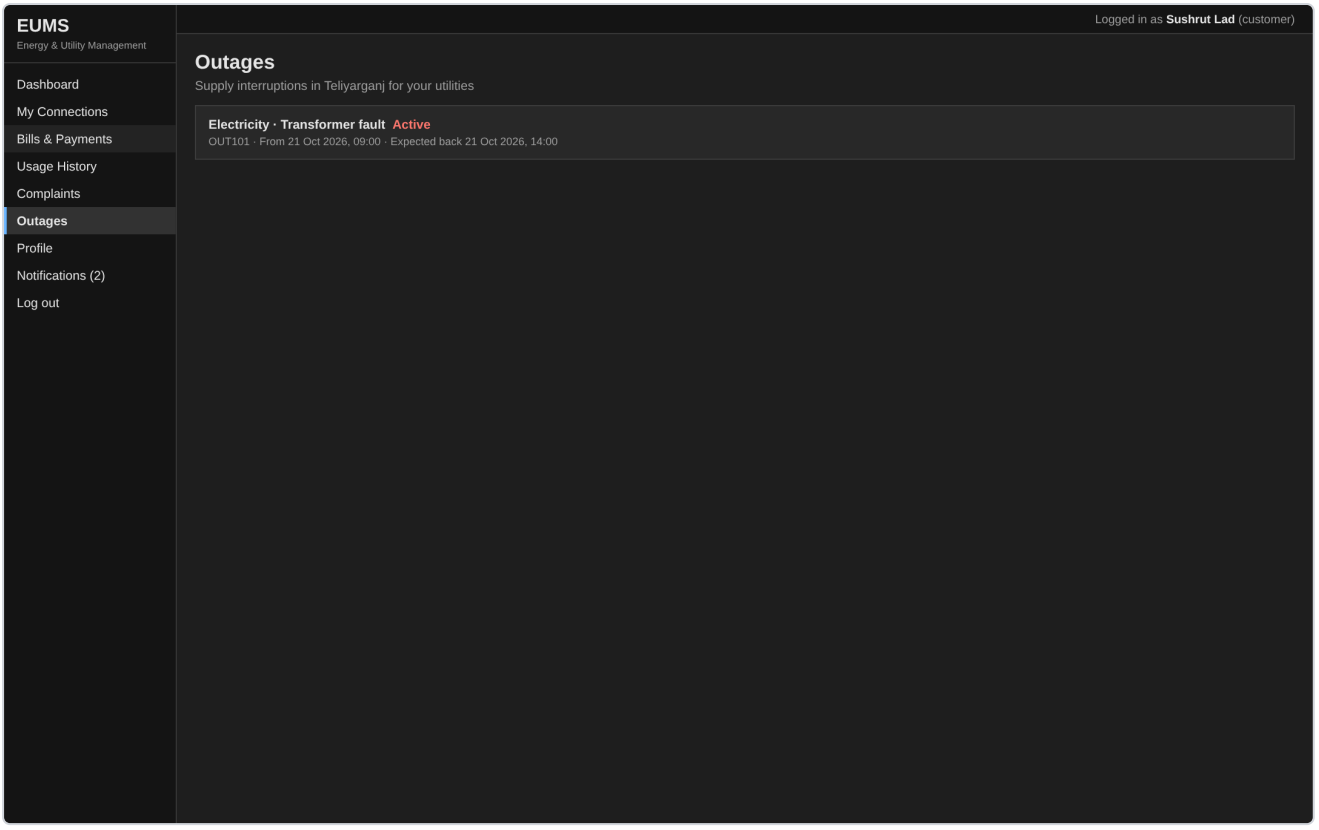


Figure 23: Customer: outages in my area

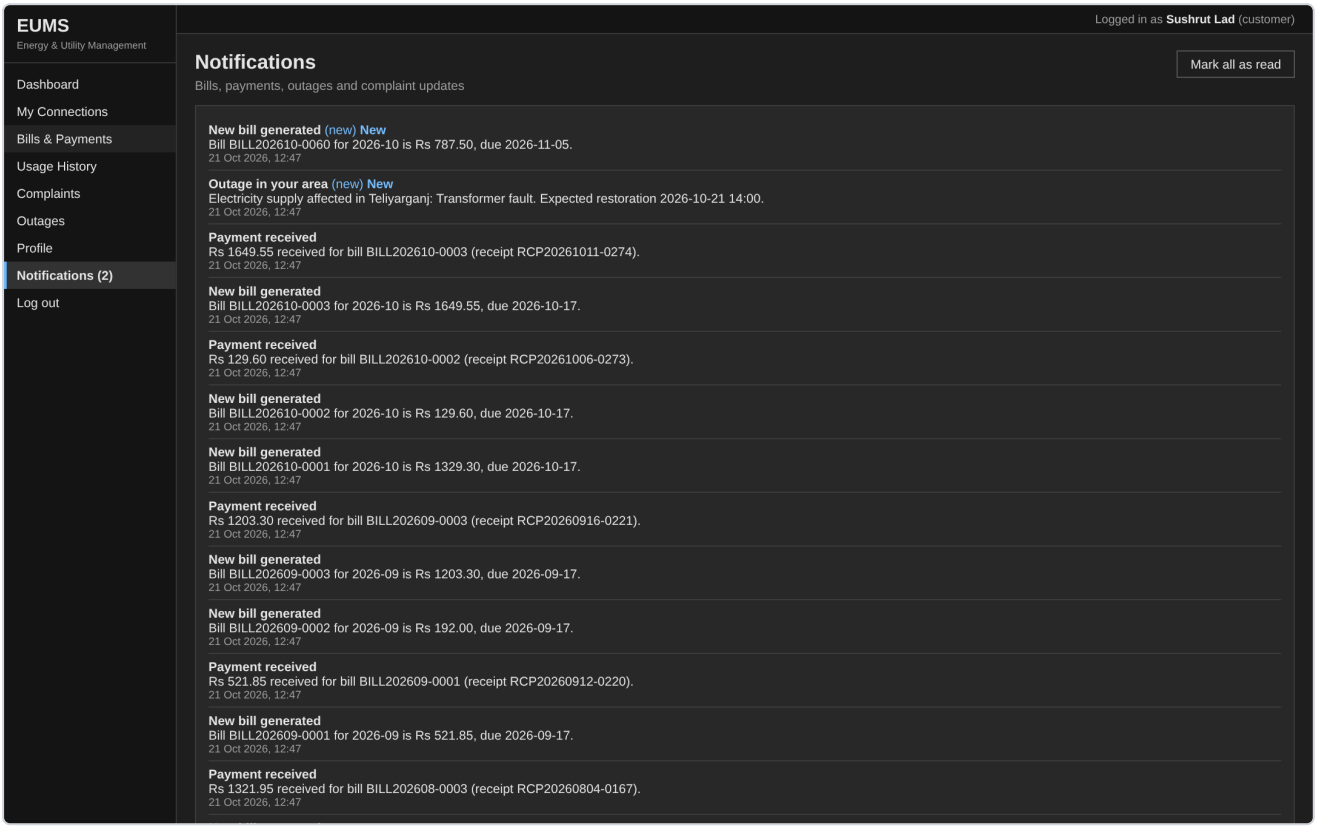


Figure 24: Customer: notifications

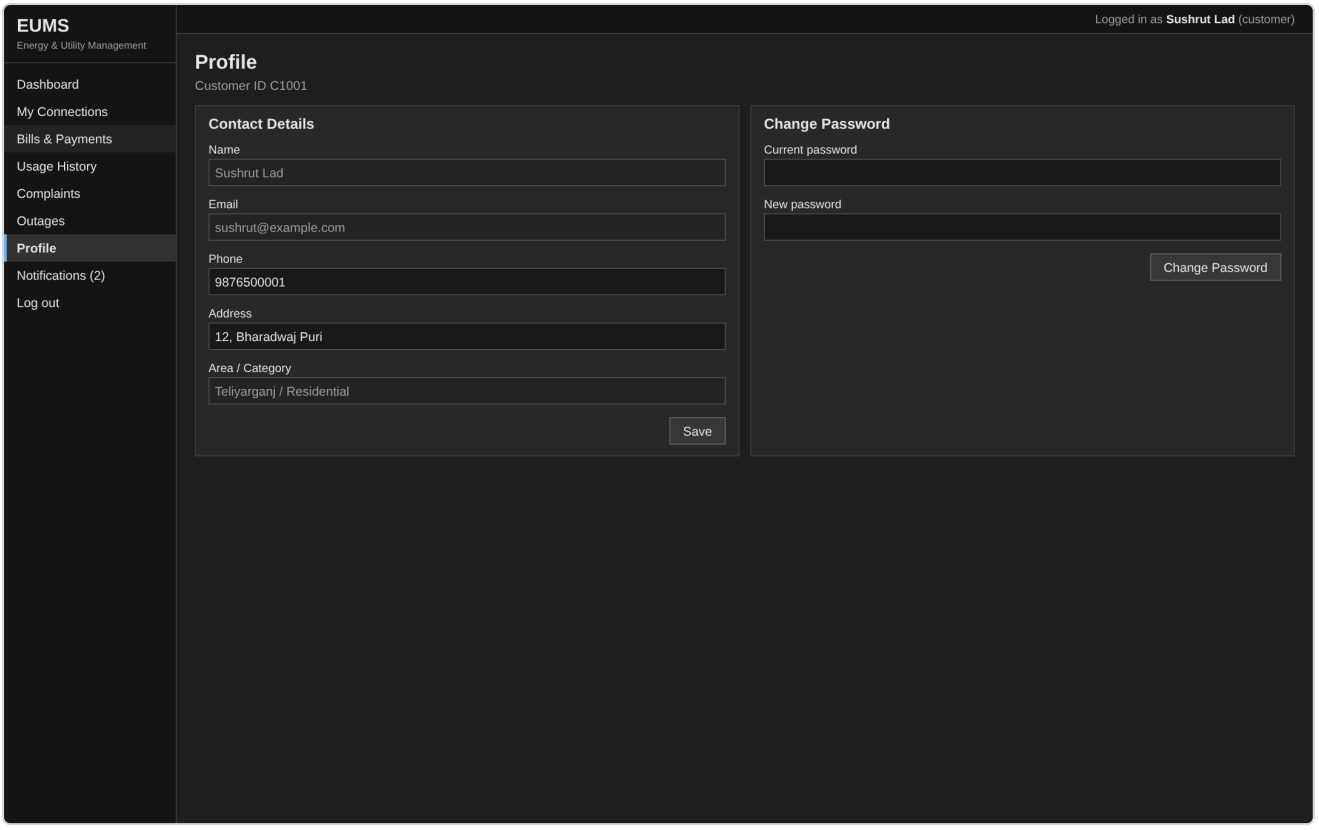


Figure 25: Customer: profile and password